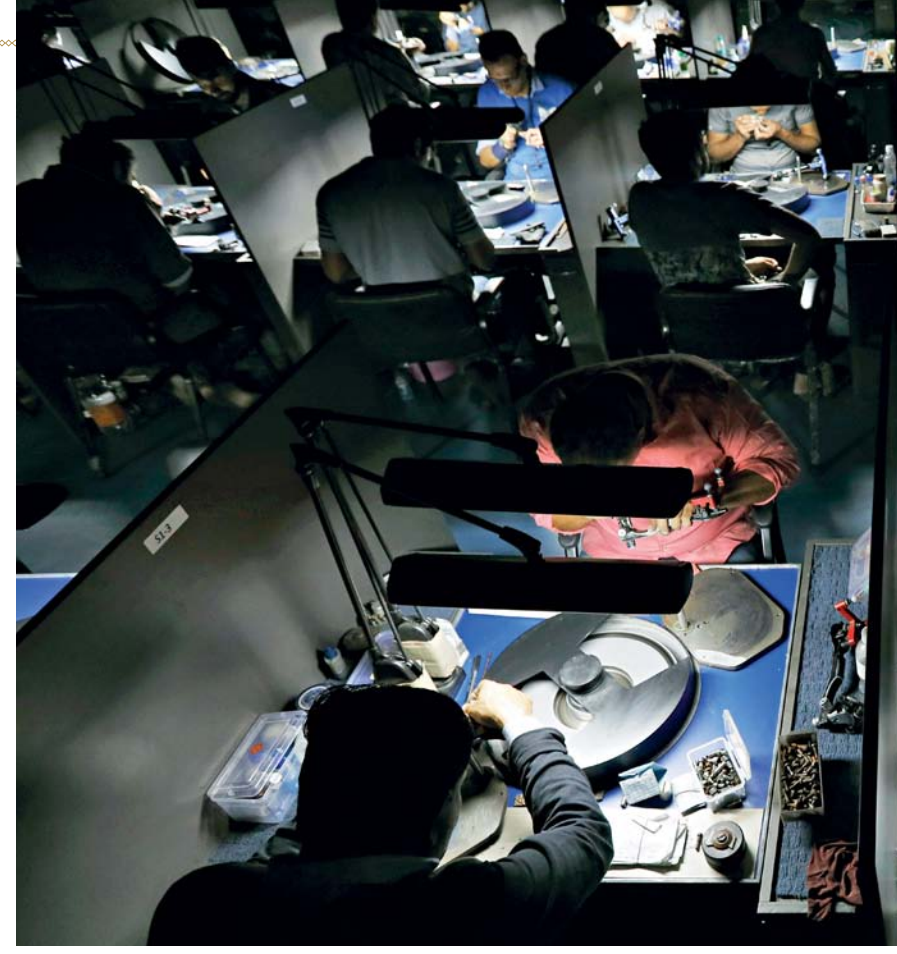


a relatively recent technology called Chemical Vapour Deposition or CVD, in which the diamond seed is placed in a chamber filled with carbon-rich gas. The chamber is then heated to temperatures upwards of 800°C, causing the gas to break down and release carbon atoms. Within weeks, the end product—a rough gemstone—is ready to be cut and polished.

Over the past decade, LGDs have gained legitimacy both in the industry and with consumers, as GIA and IGI began endorsing synthetic gems, and prominent companies such as De Beers, Pandora and Signet started offering them alongside natural diamonds. Today, over 75 per cent of jewellery stores in the US offer LGD jewellery, up from just a third in 2020, according to a January 2023 report by diamond industry analyst Edahn Golan.

HOME-GROWN LGDs

The trend has caught on in India as well. Surat, where 90 per cent of the world's mined diamonds are polished, has emerged as the natural choice for LGD manufacturers. Dinesh Navadia, the regional chairman of the Gems and Jewellery Export Promotion Council (GJEPC) in Gujarat, says over two dozen companies in the city have invested in LGD production. A few remaining labs are spread across cities like Mumbai, Delhi and Jaipur.

Lab diamonds are beginning to make an appearance in Indian jewellery stores as well. Start-ups like the Mumbai-based Itara and Coimbatore's Wondr Diamonds are launching what they are calling "earth-friendly" LGD jewellery, to cater to woke millennials who are rejecting mined diamonds on ethical grounds.

The industry in India, says Shashikant Dalichand, chairperson of the Lab Grown Diamond and Jewellery Promotion Council, has come up in a big way in the past three years. Today, the country accounts for about 15 per cent of the gem-quality LGDs cultivated globally, second only to China, according to industry estimates. Since 2017-18, India's LGD exports have surged at a CAGR of nearly 45 per cent, figures made available by GJEPC reveal. India had the biggest share in the $2.74 billion (Rs 22,400 crore at the current exchange rate) global exports of cut and polished LGDs in 2021.



"TODAY, DO YOU EVEN ASK IF THE PEARL YOU BUY COMES FROM THE OCEAN OR A FARM? LAB DIAMONDS WILL FIND ACCEPTANCE THE SAME WAY"

MUKESH PATEL
Chairman, Greenlab Diamonds

And the future seems only brighter. The LGD market in India is expected to rise to $5 billion (Rs 41,000 crore) by 2025 and reach $15 billion (Rs 1.23 lakh crore) by 2035, states a recent release issued by the Union ministry of commerce and industry. It was perhaps sensing this potential that the Union government abolished the 5 per cent customs duty on the imports of diamond seed, and sanctioned a grant of Rs 243 crore to the Indian Institute of Technology (IIT), Madras, to conduct research on producing indigenous "machinery, seeds and recipe" for LGDs. Currently, the seeds and machines are sourced from Germany, Singapore, China, Japan, the US and Russia. Indigenous technology, Navadia feels, to develop seeds and improve the manufacturing process will eliminate India's dependence on imports and save it from the vagaries of the global markets.

"LGD," the Union finance minister Nirmala Sitharaman also said in her budget speech this year, "is a technology- and innovation-driven emerging sector with high employment potential." Overall, the gems and jewellery sector in India employs about 4 million skilled